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Lesson Notes

MODULE 4 | LESSON 2

BLOCK MAXIMA: INDONESIAN GOLD

Reading Time	40 minutes
Prior Knowledge	Basic Extreme Value Theory, Value at Risk
Keywords	Block Maxima, Generalized Extreme Value (GEV) distribution

In the last lesson, we examined both approaches belonging to extreme value theory and used one of them in combination with the GARCH model to improve our VaR estimate. We do a similar analysis in this lesson, but we use the Block Maxima approach instead of the Peaks over Threshold approach.

1. Standardized Generalized Extreme Value (GEV) vs. Standardized Normal

The variable of interest can be standardized by subtracting the location parameter (which Pratiwi et al. do not define as such, but they refer to it as μ) and dividing by the scale parameter (Pratiwi et al. 3). This standardization transformation should appear eerily similar to the standardization transformation performed on Normal variables to make them standard Normal; however, though the Greek letters are the same in the formula, they mean very different things:

- μ is the mean in the Normal distribution, but the location parameter in the GEV.
- σ is the volatility in the Normal distribution, but the scale parameter in the GEV. Another interesting parallel with respect to this parameter is that in both cases it must be positive: not just non-negative because if σ is equal to zero in the Normal framework, you no longer have a Normal distribution (but rather a Dirac delta function centered around μ , where basically all of the probability is associated with this value, leaving zero probability to the rest of the distribution).

As a side note and gentle reminder, ξ does not play a part in either standardization; it is not ordinarily considered a relevant parameter in relation to the Normal distribution, but it plays the important role of the shape parameter in the GEV—which, as you will read, determines which member of the GEV family of distributions a CDF belongs to (Pratiwi et al. 3).

2. Exploring Block Maxima with Python

Now let's use Steven's GitHub repo to experiment. Starting with "2_block_maxima.ipynb," note that the first code block is just to generate some simulated sample data: `Series` contains the sum of

Normal and Pareto random variables, with parameters that you can read in the code.

You might choose, if it helps for your own clarity and for use in this notebook only, to comment out the Pareto term, like so:

```
series = pd.Series(norm.rvs(scale=NORMAL_STD, size=N_DATAPOINTS),  
  
    # + pareto.rvs(PARETO_SHAPE, size=N_DATAPOINTS),  
  
    name=EXAMPLE_NAME  
  
)
```

Above that part of the line but in the same code block, you might make these two modifications:

```
NORMAL_STD = 1.0 # standard deviation of the normal distribution  
  
#PARETO_SHAPE = 2.5 # shape parameter of the Pareto distribution
```

It's a trivial change but you might find that by changing the standard deviation to 1.0, you can more easily exploit your familiarity with the Standard Normal distribution. Commenting out the shape variable assignment is unnecessary but further illustrates that, consistent with commenting out the Pareto random variable from the "sum" in `series`, we are absolutely not explicitly incorporating the Pareto distribution here.

Now see what happens when you change the number of data points (`N_DATAPOINTS`) and the block size (`N_SAMPLES_PER_BLOCK`). For example, when we leave the defaults for these values and the seed, but make the tweaks mentioned above to ensure the observations come from a Standard Normal distribution:

```
N_DATAPOINTS = 10000  
  
N_SAMPLES_PER_BLOCK = 500
```

We get the following index-value pairs:

```
494 2.696224  
943 2.759355  
1054 3.170975  
1816 2.620574  
2060 2.929096
```

2861 2.337410
 3118 3.801660
 3577 2.662727
 4443 2.925250
 4909 2.979976
 5493 2.495995
 5656 2.841767
 6080 2.723573
 6788 2.956399
 7282 3.292694
 7756 3.094980
 8122 3.019670
 8540 2.615181
 9350 2.734218
 9901 2.865204

Let's make just two observations about the data from this distribution:

1. Each of the maxima comes from a block of 500, per the parameter settings just discussed. Clearly, the index for the first pair (494) is between 0 and 499; the second index (943) is between 500 and 999; and 1054 is between 1000 and 1499. The point here is just to emphasize how relevant data are selected in the Block Maxima Method.
2. The maximum for each block of 500 is around 3.0 and in the range of 2.3 to 3.8. This seems right, intuitively, based on what we know about the CDF for the Normal distribution. You might know by heart (or consider committing to your memory) that the 99th percentile of a Standard Normal distribution corresponds to about 2.33 standard deviations, so the fact that all our maxima are above this level accords with our expectations. We can go further: The largest value in a set of 500 could be said to correspond to the 99.8th percentile, for which we can quickly find the corresponding number of standard deviations from the mean this represents in a Standard Normal distribution.

The result is 2.88, which is in line with our rough observation above that the maxima are mostly close to 3.0. That was just a rough "eyeball" estimate (meaning a best guess, without any calculations), but it was not too far off.

What do you expect to happen if we increase the number of observations (as well as the block size so that we get back the same number of maxima for straightforward comparison), as per below?

`N_DATAPOINTS = 1000000`

```
N_SAMPLES_PER_BLOCK = 50000
```

The maxima returned are

```
22988 3.831790
60333 4.241772
118852 4.285856
182954 4.037848
216011 4.101105
299197 4.055717
341474 4.371967
358011 4.224077
448898 4.042526
457340 4.466612
527063 4.384171
567910 4.576634
632295 4.283561
679614 4.716662
703734 4.181421
786590 3.927907
807981 4.130497
857357 4.279512
932941 4.437753
985868 4.698708
```

The maximum from each consecutive set of 50,000 samples are (with one exception) above 4.0. The 99.998th percentile corresponds to 4.11, so this should not come as a surprise. In fact, nothing extraordinary is happening here at all; we are just building our insight into the nature of Block Maxima and how it relates to what we already know.

We can easily estimate parameters with confidence intervals again using Steven's code—with which you should be very familiar as it is a required reading:

```
from evt.estimated.gevmle import GEVMLE

mle = GEVMLE(block_maxima)
```

```
tail_estimate, loc_estimate, scale_estimate = mle.estimate()
```

(Excerpted from Steven)

Having performed these estimations, we display the results with Steven's code:

```
print('Tail: ', tail_estimate)

print('Loc: ', loc_estimate)

print('Scale:', scale_estimate)
```

Below, the results have been simply truncated:

Tail: Estimate(estimate=-0.2181, ci_lower=-0.5793, ci_upper=0.1429, confidence_level=0.95)

Loc: Estimate(estimate=4.1751, ci_lower=4.0653, ci_upper=4.2849, confidence_level=0.95)

Scale: Estimate(estimate=0.2204, ci_lower=0.1414, ci_upper=0.2994, confidence_level=0.95)

The estimate for the location parameter μ is about 4.175. It's no coincidence that this value is near—but greater than—the 4.11 value we calculated above for the 99.998th percentile of the Standard Normal distribution. Recall from the beginning of this lesson that μ (the location parameter) in the Gumbel distribution should not be confused with the μ (the mean) in the Normal distribution. Actually, the Gumbel location parameter represents the mode of the distribution, which is a centrality measure. The other two measures of a distribution's centrality are the mean and the median. Since the Block Maxima method is only concerned with the highest value in each block, there is a bias in the positive direction that results in the mode of 4.175 being higher than the expectation for each Normally distributed block, 4.11.

Finally, noting the positive skew of the Gumbel distribution in figure 1, what does this tell you about the mean of the Gumbel distribution? You should be able to reason that the mean of the Gumbel distribution should be higher than the mode because of the higher density of values larger than the mode than the density of values less than the mode.

3. The Definition of VaR

The English syntax in the article by Pratiwi et al. is not perfect, but the ideas are understandable, perhaps even more understandable than much of the research written in *perfect* English but with too much academic jargon. That being said, certain definitions and descriptions of Value at Risk (VaR) provided by Pratiwi et al. are sloppy and misleading, e.g.:

- "VaR measures the maximum loss in value of assets/portfolio over a predetermined time for given internal confidence" (2).
- "Value at risk (VaR) measures the maximum loss in value of an asset/portfolio over a predetermined time for a given confidence level" (7).

What is wrong with these explanations, and why are they dangerously misleading? Please discuss the problems with these definitions in the forum.

Also discuss this hypothetical: If a researcher found that the Expected Shortfall (ES) was 3% and they stated that the maximum loss is 3%, would this statement be correct? Why or why not?

4. Conclusion

In this lesson, we used the other EVT approach, in combination with GARCH, to find an improved VaR estimate. In the next lesson, we use Peaks over Threshold again, but in the context of the Ghanaian Stock Exchange. We discuss the theorem that gives rise to EVT. We also discuss the many aspects of political risk and how some researchers have attempted to incorporate political risk into their GARCH models.

References

- Pratiwi, Noviana et al. "Generalized Extreme Value Distribution for Value at Risk Analysis on Gold Price." *Journal of Physics: Conference Series*, vol. 1217, 2019, <https://iopscience.iop.org/article/10.1088/1742-6596/1217/1/012090/meta>.
- Steven. "EVT." GitHub. <https://github.com/spmvg/https://github.com/spmvg/evt>